INCH-POUND

MIL-F-70807 (AR)
AMENDMENT 5
19 September 1991
SUPERSEDING
AMENDMENT 4
15 March 1991

MILITARY SPECIFICATION

FUZE, PD: M745
(LESS BOOSTER AND LEAD)

This amendment forms a part of Military Specification MIL-F-70807 (AR), dated 11 March 1988, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2, DRAWINGS, Add "9347415 - Packing & Marking of Box Fiberboard for Fuze, Multi-Option, M734 and Fuze, PD, M745" and "9381685 - Packing & Marking of Container, M548 for Fuze, Multi-Option, M734 and Fuze, PD, M745."

Delete old drawing number and substitute new drawing number wherever it appears in this specification as follows:

Old Dwg. No.	New Dwg. No.
11737001	11927103
11737000	11927102
11737007	11927120
11744131	11927094
11740827	11927066
11740828	11927087
11740824	11927137
11744134	11927095
11744135	11927116
11744136	11927117
11744145	11927118
11737030	11927109
11737031	11927125
11737029	11927108
11744139	11927141
11744140	11927142
11744133	11927140
11744152	11927146
11737024	11927124

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PAGE 2

Old Dwg. No.	New Dwg. No.
11740823	11927136
11744129	11927093
11744128	11927092
11744147	11927144
11744146	11927143
11744126	11927139
11737016	11927122
11740821	11927084
11744127	11927091
11737012	11927106
11744137	11927096
11744138	11927097
11744130	11927115
11744141	11927098
11737020	11927079
11737021	11927080
11737027	11927081
11740819	11927134
11744125	11927138
11737013	11927121
11740806	11927130
11744121	11927067
11737010	11927105
11737026	11927107
11737016	11927122
11744148	11927074
11737023	11927123
11744151	11927145
11737000	11927102
11740826	11927086
11740828	11927087
11740829	11927088
11744120	11927090
11737030	11927109
11737001	11927103

PAGE 3

3.1.1: Change "458" to read "436," and change "428" to read "406."

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PAGE 4

3.1.4: Delete in its entirety and substitute the following:

"3.1.4 Ballistic sample.

3.1.4.1 Initial production lot. Select at random ballistic lot acceptance samples from first production lot. Ballistic lot acceptance test plan is shown in figure 2.

3.1.4.2 Production lot. Select at random ballistic lot acceptance samples from each production lot. Ballistic lot acceptance test plan is shown in figure 3."

3.2: Change "(Figure 3)" to read "(Figure 4)."

PAGE 5

3.3: Change "-25°F" to read "-50°F"

PAGE 7

3.3.2.4: Line 4, Change "low" to read "low (-50°F)"

PAGE 9

4.4.1.d Delete in its entirety and substitute the following:

"The lot size shall be a maximum of 50,000 fuzes or one month's production, whichever is smaller. Minimum lot size shall be 1,000 fuzes for lots 1 and 2 and 5,000 fuzes thereafter."

PAGE 10

4.4.3.1: Change "Figure 2" to read Figure 2 and 3."

4.4.3.2: Change "Figure 2" to read "Figure 2 and 3."

4.4.3.2: Delete the last sentence of the paragraph.

PAGE 13

4.5.1.1.2.1: Delete the words "high explosive of the."

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PAGE 15

4.5.1.2.1.5: Change "Test 101.3, Procedure I" to read "Test 101.2, Procedure II."

4.5.1.2.1.6: Change "Test 102.2" to read "Test 102.1."

4.5.1.2.1.7: Change "Test 103.2" to read "Test 103.1."

4.5.2.1: Change "The S&A Device shall be loaded with a conductive dummy detonator and placed in the tester of 4.4.4.1.2." to "The S&A Device shall be placed in the tester of 4.4.4.1.2."

PAGE 17

6.2.2: Delete last sentence of paragraph.

PAGE 21

Table I: Delete "11740827 Pad, Booster 10"
"11740828 Cap, Booster 10"

PAGE 26

* 4.4.2.3: Major 101, change "Missing stab primer spring ...0.40% ....3.1...Probe/IG test" to read "Missing stab primer spring...100%...3.1...Probe/IG test"

PAGE 27

4.4.2.4: Major 102, Change ".040" to read ".042."

PAGE 31

* 4.4.2.8, Major 101, delete in its entirety

Add Major 103, 104 as follows:

"103, .027 length out of tolerance....0.40%..3.1..Gage

104, Firing pin not perpendicular to datum D within 0.003 at max. material condition ....0.40%...3.1...Gage".

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PAGE 33

* 4.4.2.10, Add Major 104 as follows:

"104, Improper perpendicularity of pin diameter...0.40%.... 3.1...Gage"

PAGE 34

4.4.2.11: Minor 201, Change ".264" to read ".268."

PAGE 36

4.4.2.13: Change Note A:

From: "Two-thirds of the sample shall be conditioned to -50°F and one-third to +145°F for two hours minimum. The barrier shall be assembled into a loaded S&A (11737010). The S&A explosives shall be initiated electrically with the rotor in the safe position. There shall be no evidence of fragmentation, cracking or breakthrough of the barrier, bulging in excess of .015 inch on booster side face, or scorching of lead charge explosives after the M61 detonator has fire."

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TO: "Two-thirds of the sample shall be conditioned to -50°F and one-third to +145°F for two hours minimum. The barrier shall be assembled into a loaded S&A (11737010). The S&A explosives shall be initiated mechanically by stabbing the delay primer with the rotor in the safe position. There shall be no evidence of fragmentation, cracking or break-through of the barrier, bulging in excess of .015 inch on booster side face, or scorching of lead charge explosives after the M61 detonator has fired."

Note A, Add: "Special inspection level S-4 shall be applied for selecting sample size."

PAGE 46

4.4.2.22: Delete paragraph in its entirety and substitute new paragraph included.

PAGE 55

4.4.2.31: Delete Major 101 in its entirety and substitute "None defined."

PAGE 64

Delete in its entirety and substitute included pages 64 and 64a.

PAGE 64a

Change Duds "3/4" to "6/7"

PAGE 65

Delete in its entirety and substitute per included sheet.

PAGE 66

Delete in its entirety and substitute per included sheet.

PAGE 67

Figure 4: Add new figure per included sheet.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-70807 (AR)
AMENDMENT 5

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.22	CAM				11744137
					NEXT HIGHER ASSEMBLY
					11737016
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Insufficient strength of arm		0.40%	3.1	Force test (Dwg. Note 5)
102	.203 arm height or perpendicularity out-of-tolerance		0.40%	3.1	Gage
103	.140 arm length out-of-tolerance		0.40%	3.1	Gage
Minor					
201	.125 diameter out-of-tolerance		0.65%	3.1	Gage
202	.500 length out-of-tolerance		0.65%	3.1	Gage
203	Evidence of poor workmanship		0.65%	3.5	Visual

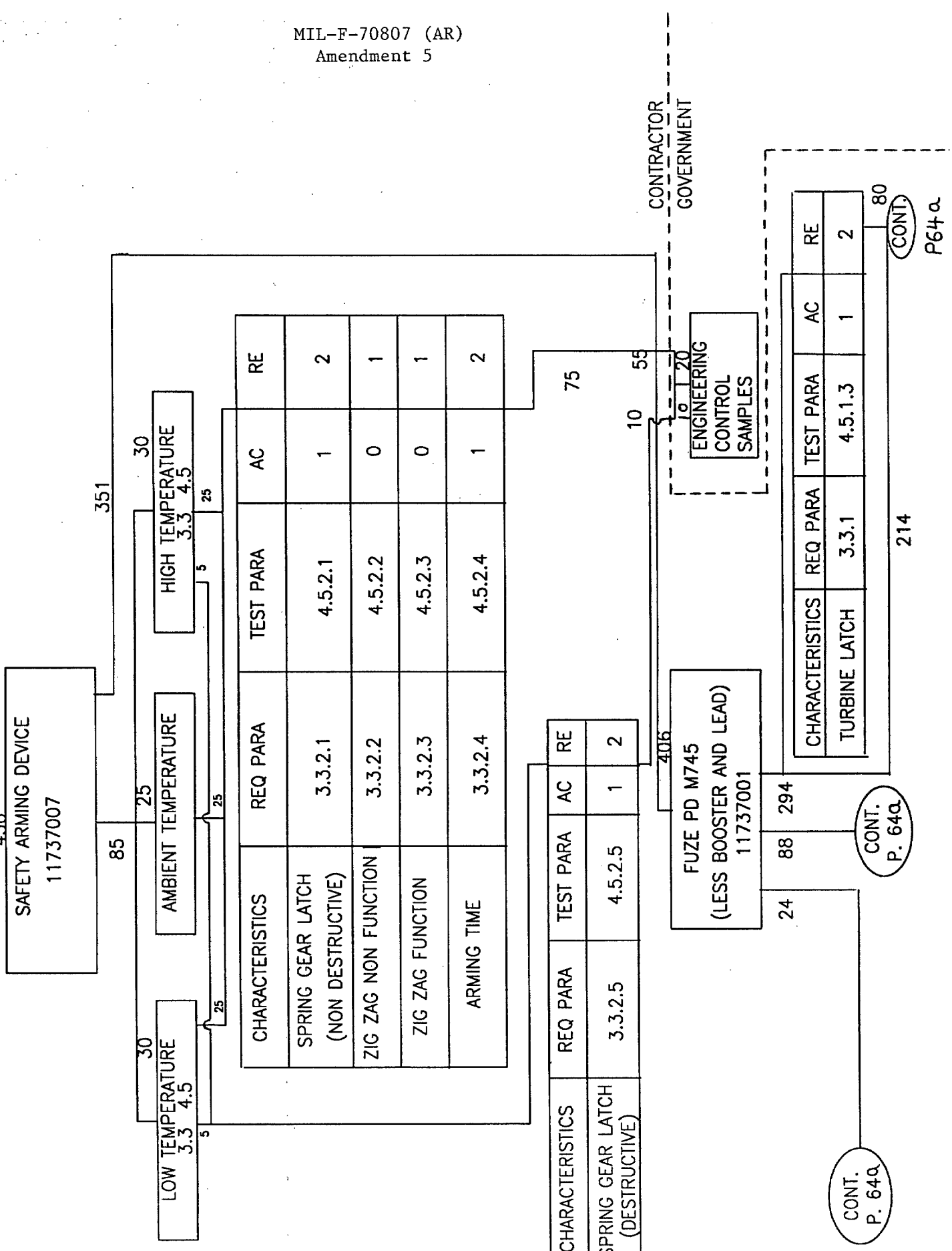
NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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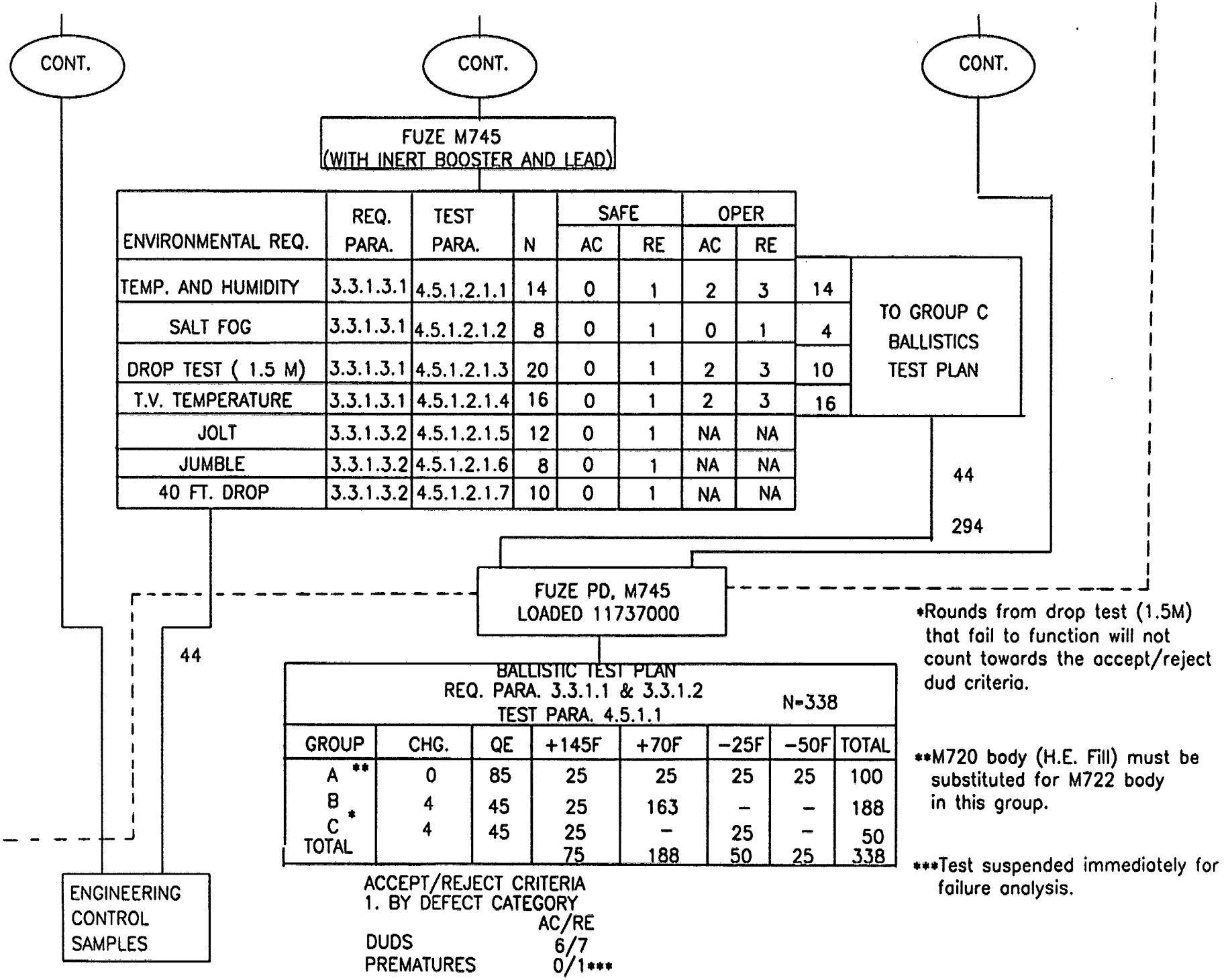
FIGURE 1

FIRST ARTICLE INSPECTION FLOW CHART



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FIGURE 1 (Continuation)



ENVIRONMENTAL REQ.	REQ. PARA.	TEST PARA.	N	SAFE AC	SAFE RE	OPER AC	OPER RE	
TEMP. AND HUMIDITY	3.3.1.3.1	4.5.1.2.1.1	14	0	1	2	3	14
SALT FOG	3.3.1.3.1	4.5.1.2.1.2	8	0	1	0	1	4
DROP TEST (1.5 M)	3.3.1.3.1	4.5.1.2.1.3	20	0	1	2	3	10
T.V. TEMPERATURE	3.3.1.3.1	4.5.1.2.1.4	16	0	1	2	3	16
JOLT	3.3.1.3.2	4.5.1.2.1.5	12	0	1	NA	NA	
JUMBLE	3.3.1.3.2	4.5.1.2.1.6	8	0	1	NA	NA	
40 FT. DROP	3.3.1.3.2	4.5.1.2.1.7	10	0	1	NA	NA	

BALLISTIC TEST PLAN
REQ. PARA. 3.3.1.1 & 3.3.1.2
TEST PARA. 4.5.1.1
N=338

GROUP	CHG.	QE	+145F	+70F	-25F	-50F	TOTAL
A **	0	85	25	25	25	25	100
B *	4	45	25	163	–	–	188
C *	4	45	25	–	25	–	50
TOTAL			75	188	50	25	338

ACCEPT/REJECT CRITERIA
1. BY DEFECT CATEGORY

	AC/RE
DUDS	6/7
PREMATURES	0/1***

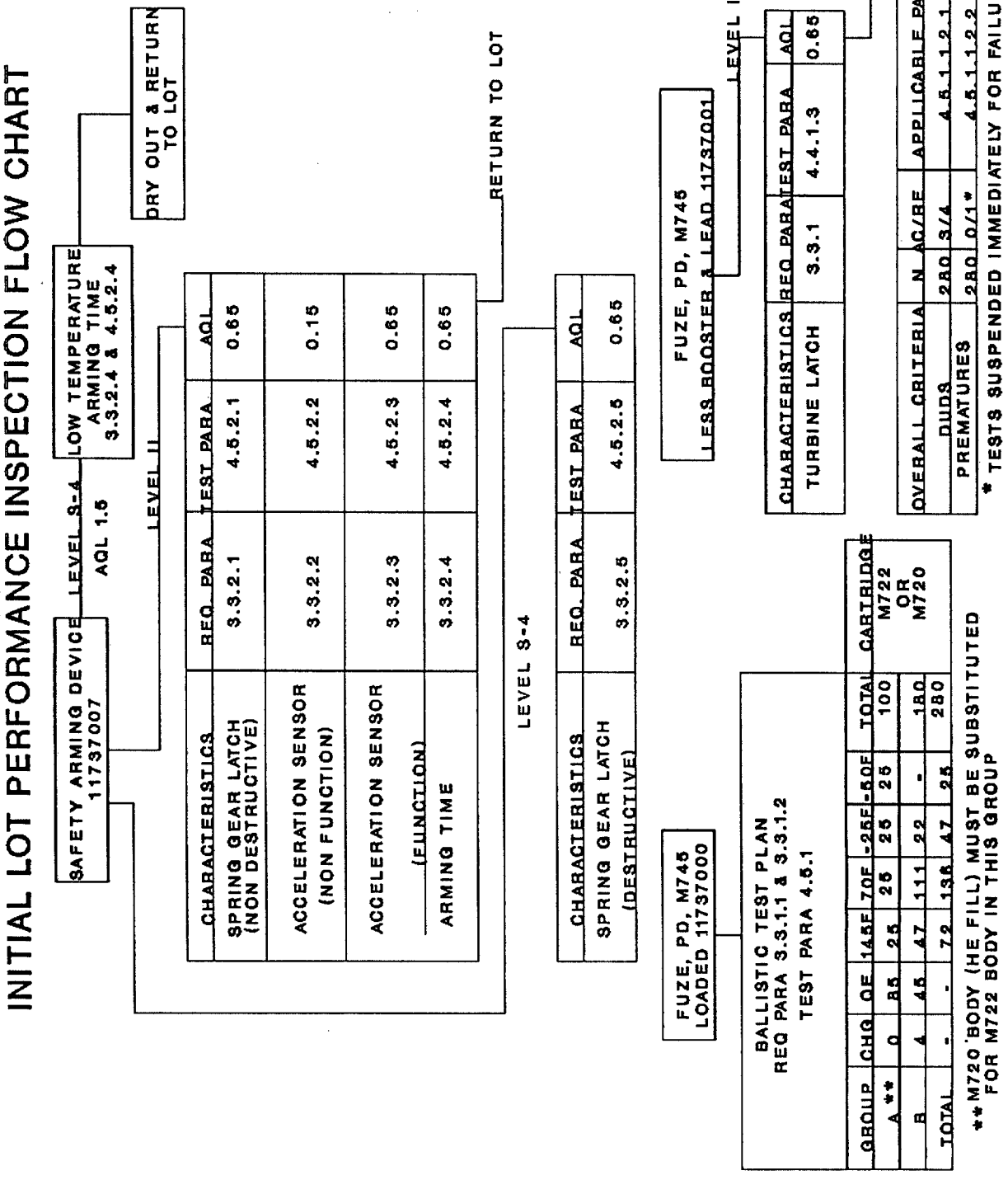
*Rounds from drop test (1.5M) that fail to function will not count towards the accept/reject dud criteria.

**M720 body (H.E. Fill) must be substituted for M722 body in this group.

***Test suspended immediately for failure analysis.

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FIGURE 2
INITIAL LOT PERFORMANCE INSPECTION FLOW CHART



CHARACTERISTICS	REQ. PARA	TEST PARA	AQL
SPRING GEAR LATCH (NON DESTRUCTIVE)	3.3.2.1	4.5.2.1	0.65
ACCELERATION SENSOR (NON FUNCTION)	3.3.2.2	4.5.2.2	0.15
ACCELERATION SENSOR (FUNCTION)	3.3.2.3	4.5.2.3	0.65
ARMING TIME	3.3.2.4	4.5.2.4	0.65

CHARACTERISTICS	REQ. PARA	TEST PARA	AQL
SPRING GEAR LATCH (DESTRUCTIVE)	3.3.2.5	4.5.2.5	0.65

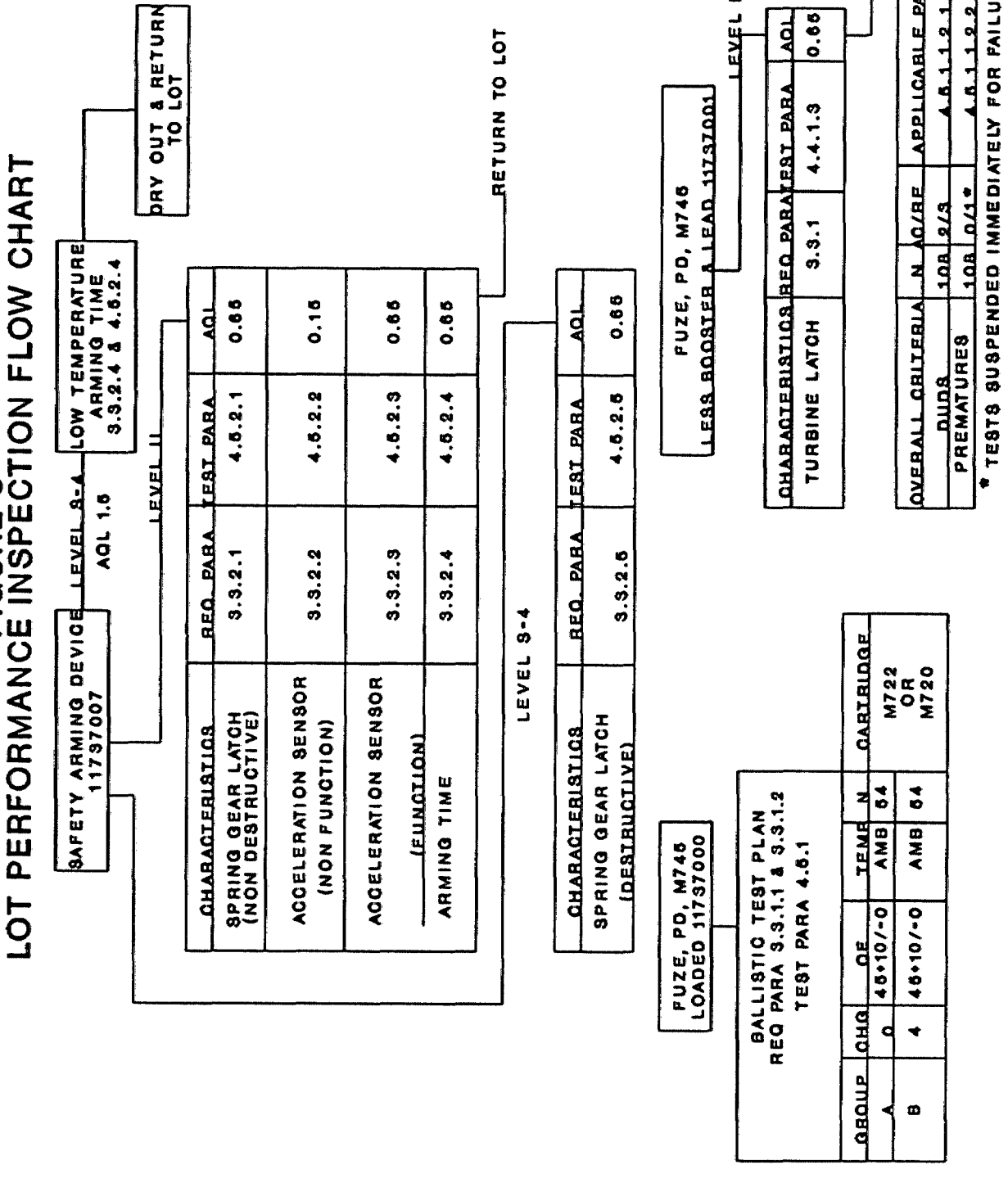
CHARACTERISTICS	REQ. PARA	TEST PARA	AQL
TURBINE LATCH	3.3.1	4.4.1.3	0.65

OVERALL CRITERIA	N	AC/RE	APPLICABLE PARA
DUDS	280	3/4	4.5.1.1.2.1
PREMATURES	280	0/1*	4.5.1.1.2.2

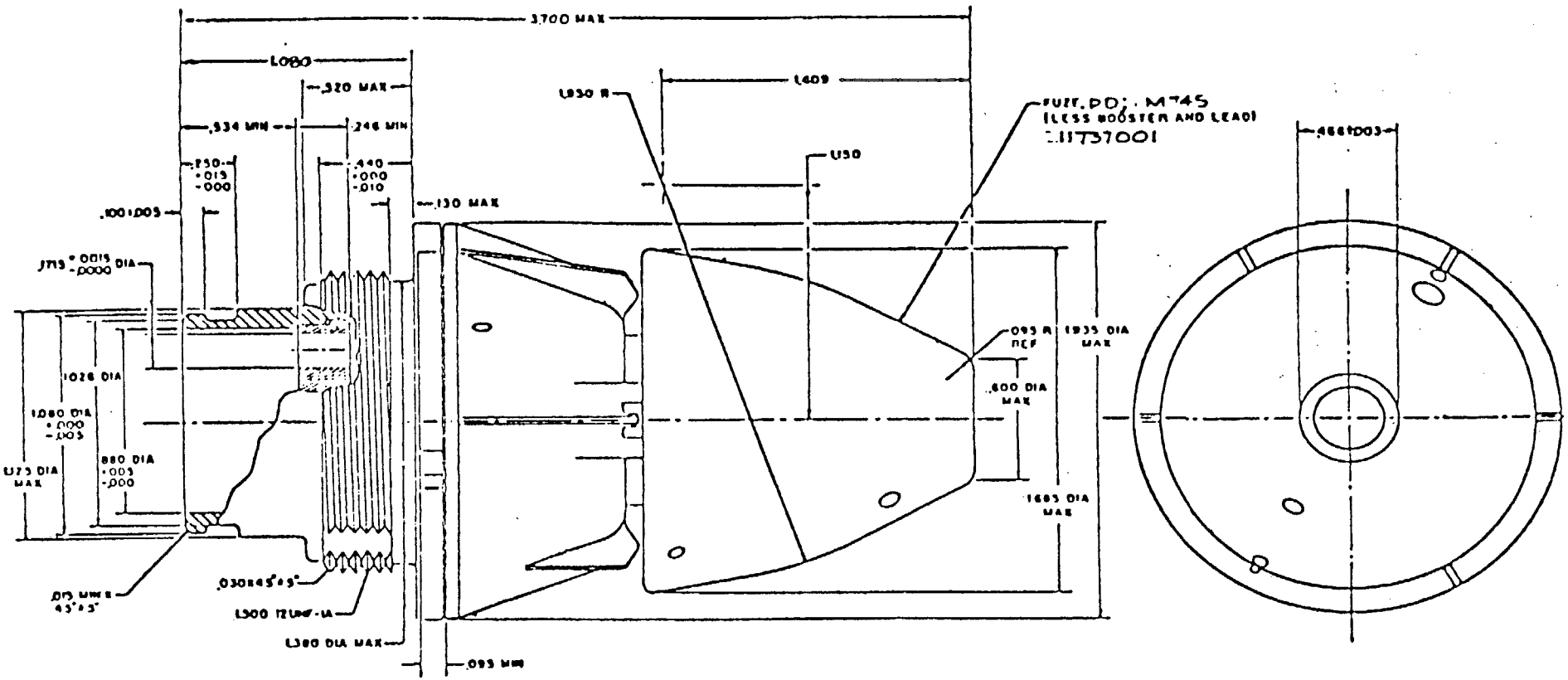
GROUP	CHG	QE	145F	70F	-25F	-50F	TOTAL	CARTRIDGE
A **	0	85	25	25	25	25	100	M722 OR M720
B	4	45	47	111	22	-	180	
TOTAL	-	-	72	136	47	25	280	

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FIGURE 3
LOT PERFORMANCE INSPECTION FLOW CHART



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NOTES

1. WEIGHT OF FULLY LOADED FUZE IS 50 + .00/-.10 LBS
2. .1715 DIA HOLE SHALL BE ADJACENT TO SCALLOP IN BOOSTER CAVITY SUCH THAT LEAD CAN BE INSERTED
3. S & A ROTOR SHALL BE IN SAFE POSITION (DWGS 11737010 AND 11737007.) M61 DETONATOR (8MS5554) IN PRIMER HOUSING ASS'Y (11737026) SHALL NOT BE VISIBLE THROUGH BARRIER LEAD HOLE IN AFT END OF FUZE.
4. FUZE SHALL BE PERMANENTLY MARKED PER DWGS 11740819
5. FUZE SHOWN IS NOT FULLY LOADED. SEE DWG NO. 11737000 FOR LOADED ASSEMBLY AND COMPLETE INTERFACE DIMENSIONS

FIGURE 4 INTERFACE CONTROL DRAWING

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